

DBML

```
Table User {  
  id integer [primary key]  
  username varchar  
  email varchar  
}
```

```
Table Loan {  
  id integer [primary key]  
  id_user integer [not null]  
  id_book integer [not null]  
  loan_date date  
}
```

```
Table Book {  
  id integer [primary key]  
  title varchar  
  author varchar  
}
```

```
Ref: Loan.id_user ?> User.id  
Ref: Loan.id_book ?> Book.id
```

```
Records User(id, username, email) {  
  0, 'Ana Gómez', 'ana@email.com'  
  1, 'Carlos Ruíz', 'carlos@email.com'  
}
```

```
Records Book(id, title, author) {  
  0, 'El Principito', 'Antoine de Saint-Exupéry'  
  1, 'Cien Años de Soledad', 'Gabriel García Márquez'  
}
```

```
Records Loan(id, id_user, id_book, loan_date) {  
  0, 0, 0, '2026-03-01'  
  1, 1, 1, '2026-03-02'  
  2, 0, 1, '2026-03-05'  
}
```

MYSQL SENTENCE

LLM PROMPT:

Convierte de DBML a una sola línea de MYSQL para ejecutar en bash después de ingresar a la shell

LLM OUTPUT:

```
CREATE TABLE User (id INT PRIMARY KEY, username VARCHAR(255), email VARCHAR(255)); CREATE TABLE Book (id INT PRIMARY KEY, title VARCHAR(255), author VARCHAR(255)); CREATE TABLE Loan (id INT PRIMARY KEY, id_user INT NOT NULL, id_book INT NOT NULL, loan_date DATE, FOREIGN KEY (id_user) REFERENCES User(id), FOREIGN KEY (id_book) REFERENCES Book(id)); INSERT INTO User (id, username, email) VALUES (0, 'Ana Gómez', 'ana@email.com'), (1, 'Carlos Ruíz', 'carlos@email.com'); INSERT INTO Book (id, title, author) VALUES (0, 'El Principito', 'Antoine de Saint-Exupéry'), (1, 'Cien Años de Soledad', 'Gabriel García Márquez'); INSERT INTO Loan (id, id_user, id_book, loan_date) VALUES (0, 0, 0, '2026-03-01'), (1, 1, 1, '2026-03-02'), (2, 0, 1, '2026-03-05');
```

DOCKER PROCESS

```
~ ) sudo docker run --name mysql-practical -e MYSQL_ROOT_PASSWORD=root -p 3306:3306 -d mysql:8
450e5af7c4664c2f61e7678af45f25778014c9a8d222fca47b85361fcf776509

~ ) sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
450e5af7c466   mysql:8   "docker-entrypoint.s..." 15 seconds ago Up 14 seconds 0.0.0.0:3306→3306/tcp, [::]:3306→3306/tcp, 33060/tcp   mysql-practical

~ ) sudo docker exec -it 450e5af7c466 bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database practical;
Query OK, 1 row affected (0.04 sec)

mysql> use practical;
Database changed
mysql>
```

PatricioReyes_DTS5_DBAdmin_Practica1

```
mysql> CREATE TABLE User (id INT PRIMARY KEY, username VARCHAR(255), email VARCHAR(255)); CREATE TABLE Book (id INT PRIMARY KEY, title VARCHAR(255), author VARCHAR(255)); CREATE TABLE Loan (id INT PRIMARY KEY, id_user INT NOT NULL, id_book INT NOT NULL, loan_date DATE, FOREIGN KEY (id_user) REFERENCES User(id), FOREIGN KEY (id_book) REFERENCES Book(id)); INSERT INTO User (id, username, email) VALUES (0, 'Ana Gmez', 'ana@email.com'), (1, 'Carlos Ruz', 'carlos@email.com'); INSERT INTO Book (id, title, author) VALUES (0, 'El Principito', 'Antoine de Saint-Exupry'), (1, 'Cien Años de Soledad', 'Gabriel García Márquez'); INSERT INTO Loan (id, id_user, id_book, loan_date) VALUES (0, 0, 0, '2026-03-01'), (1, 1, 1, '2026-03-02'), (2, 0, 1, '2026-03-05');
Query OK, 0 rows affected (0.03 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> Select l.id, u.username, u.email, b.title, b.author, l.loan_date from Loan as l Join User as u
on l.id_user = u.id Join Book as b on l.id_book;
+----+-----+-----+-----+-----+-----+
| id | username | email | title | author | loan_date |
+----+-----+-----+-----+-----+-----+
| 1 | Carlos Ruz | carlos@email.com | El Principito | Antoine de Saint-Exupry | 2026-03-02 |
| 2 | Ana Gmez | ana@email.com | El Principito | Antoine de Saint-Exupry | 2026-03-05 |
| 1 | Carlos Ruz | carlos@email.com | Cien Años de Soledad | Gabriel García Márquez | 2026-03-02 |
| 2 | Ana Gmez | ana@email.com | Cien Años de Soledad | Gabriel García Márquez | 2026-03-05 |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```